Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	OptiCrop - Smart Crop Recommendation Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal aims to help farmers choose the optimal crop using data-driven machine learning. It replaces guesswork with statistical accuracy, improving yields and resource efficiency. Key features include an ML-based crop recommendation model and a real-time prediction web app.

Project Overview	
Objective	Recommend the optimal crop for a farm using N, P, K, temperature, humidity, pH and rainfall via machine learning.
Scope	A Flask web app serving a scikit-learn model trained on 22 crops; farmers input soil/climate data and get a crop.
Problem Statement	
Description	Farmers lack data-driven, soil-specific guidance and rely on guesswork, causing low yields and wasted resources.
Impact	Better crop selection raises yield, cuts fertilizer/water waste, and improves farmer income and resource efficiency.
Proposed Solution	
Approach	Train a Logistic Regression classifier on historical agronomic data and expose it through a simple web interface.
Key Features	- ML-based crop recommendation (22 crops)

	- Real-time prediction API (~150 ms) - Glassmorphism UI with error handling - Deployed free on Vercel

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv